

# Optimization

## 2.4 Bisection method

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José Firmino Aguilar Madeira

## With $f'$ , is one point enough?

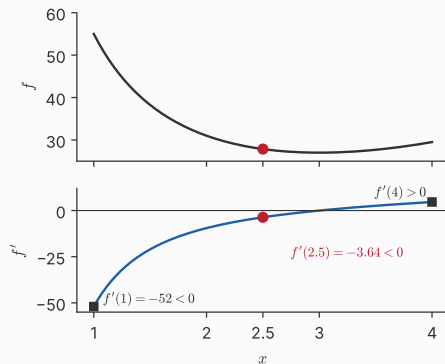
With values of  $f$  only (2.1–2.2) we needed **two** points to eliminate. With  $f'$ : at an interior minimum,  $f'(x^*) = 0$ .

Can we do the same using only the sign of  $f'$ ?

Guiding example:  $f(x) = x^2 + 54/x$  on  $[1, 4]$ ,

$f'(x) = 2x - 54/x^2$ . A single point:

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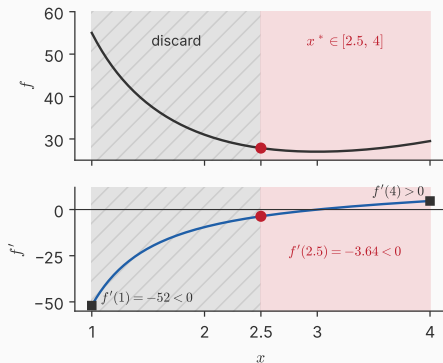
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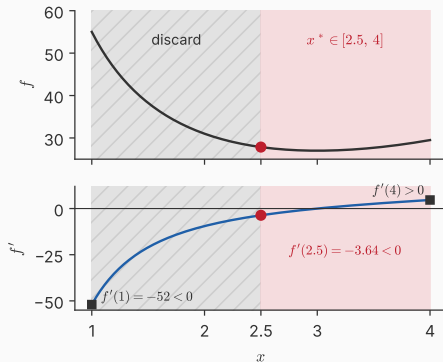
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Where to put the point? **In the middle**: whatever the sign, half remains.

sign of  $f'$  at the midpoint  $\rightarrow$  **bisection**



# Why the sign is enough

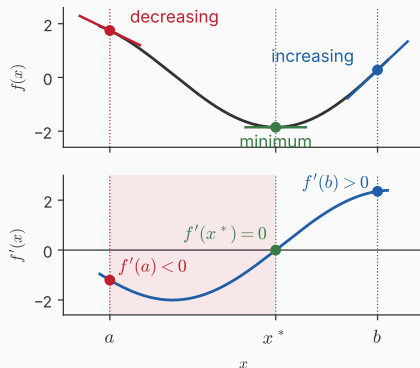
**Assumptions:**  $f'$  continuous on  $[a, b]$ ,  
 $f'(a) < 0 < f'(b)$  and **a single zero**, where  $f'$   
changes sign  $- \rightarrow +$ . Then that zero is the only  
interior minimum  $x^*$ . In particular,  $f$  strictly  
convex with  $f'(a) < 0 < f'(b)$  is enough.

The sign of  $f'$  at a point tells **on which side** the  
minimum is:

- $f'(x) < 0$ :  $f$  is still decreasing  $\Rightarrow x^* > x$ ;
- $f'(x) > 0$ :  $f$  is already increasing  $\Rightarrow x^* < x$ .

It tells *on which side*, not *where*:

$f' = -3.64$  and  $f' = -0.0001$ : the same half.



$f'(a) < 0 < f'(b)$  alone guarantees *at least one*  
zero; without uniqueness there may be several  
stationary points and we do not know whether we  
isolate the desired minimum.

# The algorithm

## Bisection (on $f'$ )

- Step 1** Choose  $a < b$  with  $f'(a) < 0 < f'(b)$  and  $\varepsilon > 0$ ; set  $x_1 = a$  and  $x_2 = b$ .
- Step 2** Set  $z = \frac{x_1 + x_2}{2}$  and compute  $f'(z)$ .
- Step 3** If  $|f'(z)| \leq \varepsilon$ , **terminate**:  $x^* \approx z$ . Otherwise, if  $f'(z) < 0$ , set  $x_1 = z$  (minimum to the right); if  $f'(z) > 0$ , set  $x_2 = z$  (minimum to the left). Go to Step 2.

**Note:**  $|f'(z)| \leq \varepsilon$  says that  $f$  is nearly flat at  $z$ , *not* that  $z$  is within  $\varepsilon$  of  $x^*$ . Variant: stop when  $x_2 - x_1 \leq \varepsilon_x$  and return the midpoint, which guarantees  $|x - x^*| \leq \varepsilon_x/2$ .

- **Order 1** information: it uses  $f'$  (its sign, to choose the half; its value, to stop), not  $f$ .
- It requires  $f'(a) < 0 < f'(b)$ : check before starting. If it fails, the minimum is not bracketed and bisection does not apply in this form.
- Cost: **one evaluation of  $f'$  per iteration**. If  $f'$  is not available analytically, it is approximated by finite differences (2.6) — two evaluations of  $f$ .

# How many iterations? And how much does each one cost?

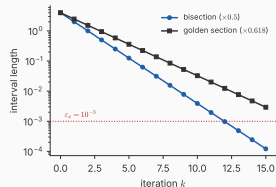
Each iteration halves the interval: after  $k$  iterations,  $x_2 - x_1 = \left(\frac{1}{2}\right)^k (b - a)$ . With  $|f'(z)| \leq \varepsilon$ , the number of iterations depends on  $f$  and is not known in advance. In the variant (stop when  $x_2 - x_1 \leq \varepsilon_x$ ; the midpoint has  $|x - x^*| \leq \varepsilon_x/2$ ), it is:

$$k = \left\lceil \log_2 \frac{b - a}{\varepsilon_x} \right\rceil$$

## Example

$[a, b] = [0, 2]$ ,  $\varepsilon_x = 10^{-3}$ :  $\log_2 2000 = 10.97$ , so  $k = 11$ . With  $\varepsilon_x = 10^{-4}$ :  $k = 15$ .

Each decimal digit costs  $\log_2 10 \approx 3.3$  iterations (golden section: 4.8).



## But what is the cost per evaluation of $f$ ?

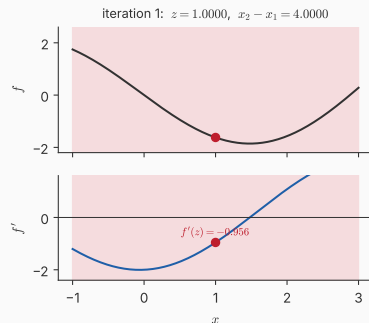
Bisection:  $\times 0.5$  per  $f'$ ; golden section:  $\times 0.618$  per new  $f$ . With  $f'$  by central differences (2 evaluations of  $f$ ):  $\sqrt{0.5} \approx 0.707$  per evaluation — **worse**.

A cost conversion, not an intrinsic rate of bisection; it assumes 2 new evaluations of  $f$  per  $f'$ , with no reuse.

# Example 1: $f(x) = -2 \sin x + x^2/16$ on $[-1, 3]$

$$f'(x) = -2 \cos x + x/8; \quad f'(-1) = -1.21 < 0, \\ f'(3) = 2.35 > 0 \checkmark; \quad \varepsilon = 0.03$$

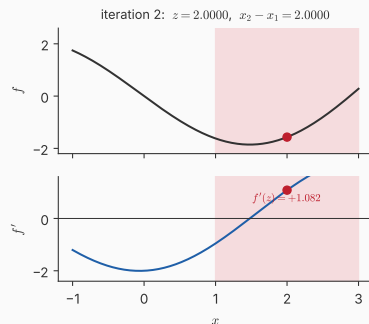
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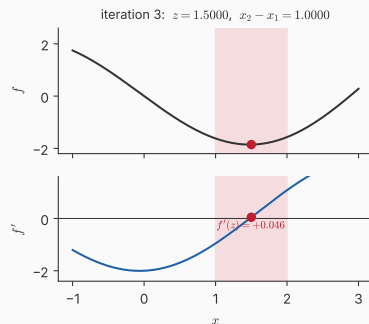
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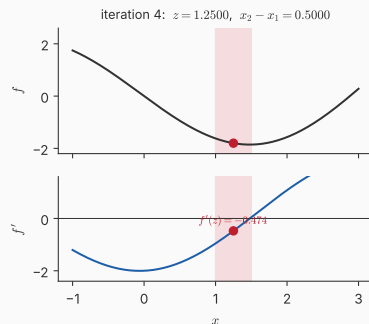
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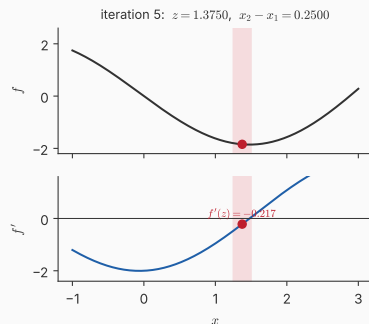
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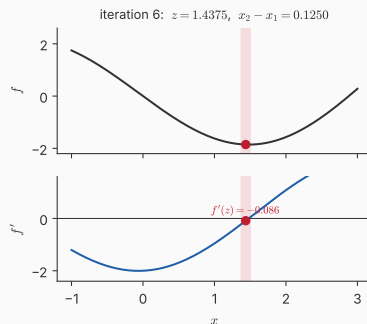
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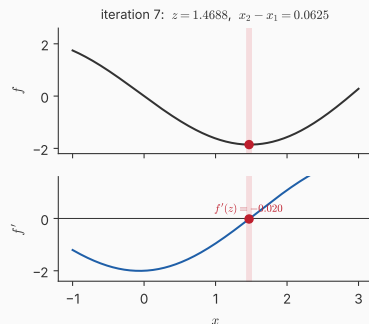
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| 7   | 1.4375  | 1.5000 | 1.4688 | -1.8548 | -0.0201 |

$|f'(z)| = 0.0201 \leq \varepsilon$ : **terminate**,  $x^* \approx z = 1.4688$  (exact: 1.4783).

With  $\varepsilon = 0.05$  it would terminate at iteration 3, at  $z = 1.5$

( $|f'(z)| = 0.0460$ ), with error 0.022.

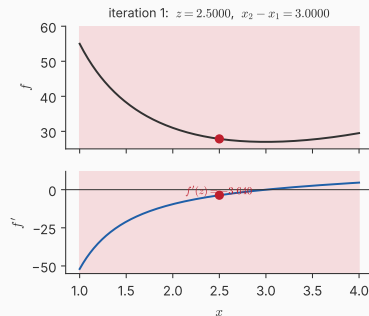


## Example 2: $f(x) = x^2 + 54/x$ on $[1, 4]$

$$f'(x) = 2x - 54/x^2; \quad f'(1) = -52 < 0, \quad f'(4) = 4.625 > 0$$

✓;  $\varepsilon = 0.05$  (the golden-section interval of 2.2)

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|-----|--------|--------|--------|---------|---------|
| 1   | 1.0000 | 4.0000 | 2.5000 | 27.8500 | -3.6400 |

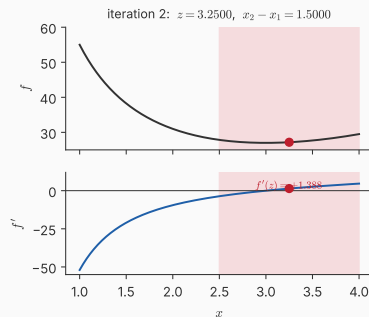


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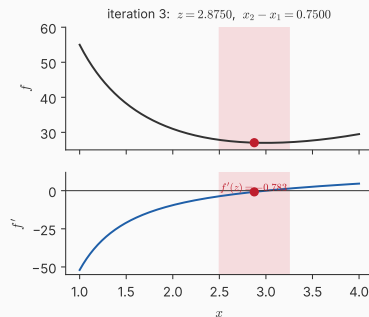


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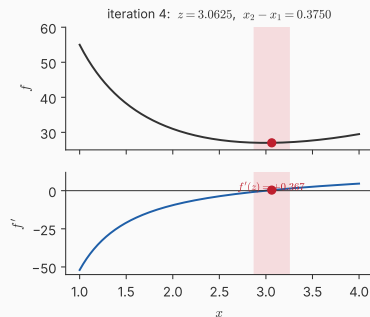


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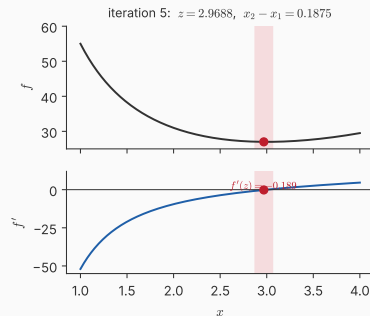


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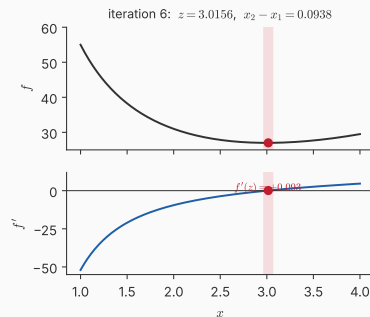


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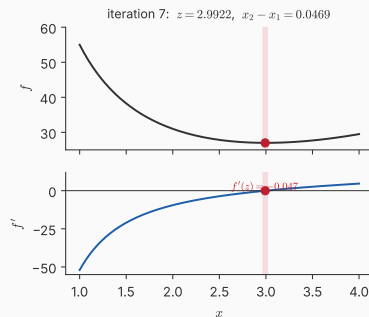
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| 5   | 2.8750 | 3.0625 | 2.9688 | 27.0030 | -0.1895 |
| 6   | 2.9688 | 3.0625 | 3.0156 | 27.0007 | +0.0933 |
| 7   | 2.9688 | 3.0156 | 2.9922 | 27.0002 | -0.0470 |

$|f'(z)| = 0.0470 \leq \varepsilon$ : **terminate**,  $x^* \approx z = 2.9922$  (exact:  $x^* = 3$ ).

Variant with 15 reductions:  $x_2 - x_1 = 3/2^{15} = 9.2 \times 10^{-5}$ ; golden section, same  $[1, 4]$ , 15 reductions:  $L = 0.0022$ , 24 times larger (but with  $f$  only). The  $f(z)$  column is for reading only: the method does not use it.



## To solve

Apply bisection with  $\varepsilon = 0.01$  and confirm:

| $f(x)$                      | interval | iterations | final $z$ | $f(z)$  |
|-----------------------------|----------|------------|-----------|---------|
| $x + \frac{31}{x+3}$        | $[0, 5]$ | 6          | 2.5781    | 8.1355  |
| $-3 \sin x + \frac{x^2}{8}$ | $[0, 4]$ | 10         | 1.4492    | -2.7153 |

- In the first, check analytically:  $f'(x) = 1 - \frac{31}{(x+3)^2} = 0 \Rightarrow x^* = \sqrt{31} - 3 = 2.5678$ . The error of  $z$  is  $0.0104 > \varepsilon$ :  $|f'(z)| \leq \varepsilon$  does not guarantee  $|z - x^*| \leq \varepsilon$ .
- In the second ( $x^* = 1.4497$ ), compare with golden section in 2.2 (1.4489,  $L_{15} = 0.0029$ ) and with the variant  $x_2 - x_1 \leq \varepsilon_x$  with 15 reductions ( $4/2^{15} = 1.2 \times 10^{-4}$ ): after 15 reductions, which one guarantees the smaller error?

# On the computer

The core, in MATLAB ( $\text{tol} = \varepsilon$ ):

```
if ~(df(a)<0 && df(b)>0), error('bracketing'), end
x1 = a; x2 = b;
while true
    z = (x1 + x2)/2; d = df(z);
    if abs(d) <= tol, break, end % finished
    if d < 0
        x1 = z; % minimum to the right
    else
        x2 = z; % minimum to the left
    end
end
x = z;
```

The full function (Bisection) is in the course companion code.

It uses only  $f'$ :  $n_g = 1$  per iteration, plus  $f'(a)$  and  $f'(b)$  to check the bracketing ( $n_g = k + 2$ );  $n_f = 0$ .

The full function also has a maximum number of iterations and, optionally, the variant  $x_2 - x_1 \leq \varepsilon_x$ .

Without an analytical  $f'$ , central differences (2.6),  $n_f = 2$  per  $f'$ :

```
h = 1e-6;
df = @(x) (f(x+h) - f(x-h)) / (2*h);
```

near the zero, the **sign** of  $f'$  may also have numerical noise — see 2.6

## KEY TAKEAWAYS — 2.4

1. For an interior minimum of a differentiable  $f$ ,  $f'(x^*) = 0$ ; under the assumptions of this section, minimizing is **equivalent** to locating that zero.
2. With  $f'$  continuous,  $f'(a) < 0 < f'(b)$  and a single zero, the **sign** at the midpoint tells which half to keep:  $f'(z) < 0 \Rightarrow x_1 = z$ ;  $f'(z) > 0 \Rightarrow x_2 = z$ . **Terminate when**  
 $|f'(z)| \leq \epsilon$ .
3.  $L_k = 2^{-k}L_0$ : linear convergence with ratio  $\frac{1}{2}$ ; in the variant  $x_2 - x_1 \leq \epsilon_x$ , the number of iterations is predictable,  $k = \lceil \log_2(L_0/\epsilon_x) \rceil$ . One evaluation of  $f'$  per iteration, none of  $f$  (with  $f'$  by central differences, golden section reduces more per evaluation of  $f$ ).
4. It chooses the half by the sign of  $f'$  only: **it knows which side the zero is on, not where**.

Next (2.5): what if we also use the *value* of  $f'(z)$  and the slope of  $f'$ , that is  $f''(z)$ , to **estimate directly where** the zero is? — the Newton–Raphson method.

**References** Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996 — Ch. 2.

## Resources

- Companion code: Bisection (stopping rule  $|f'(z)| \leq \varepsilon$ ; variant  $x_2 - x_1 \leq \varepsilon_x$ );  $f'$  by central differences (df,  $h = 10^{-6}$ ).